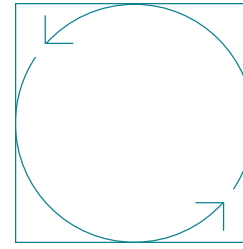


Machine Learning in Industry*

by Daniel Meister, CTO at Datahouse

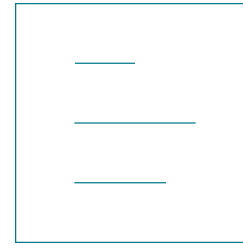


Machine Learning in Industry*

Disclaimer

* Obviously, the Data Science and Machine Learning industry is growing very fast and it would be ignorant to claim that I can give you a good overview of all the industry practices.

So the following slides just summarise a few personal thoughts on how machine learning and data science in general are used and/or should be used in the different companies and industries I'm familiar with.



Machine Learning in Industry

Agenda

What is Machine Learning / Applications

A non-exhaustive list of current machine learning applications in solving real-world problems; some examples from my own team.

General Remarks

Some general remarks about models and their business application

Ethical and Regulatory Considerations

A few words about privacy and ethics of machine learning (and working with data in general)

Daily Routine

What does a ML engineer really do in their daily work

What is Machine Learning?

What is Machine Learning?

What is Artificial Intelligence?

VOL. LIX. No. 236.]

[October, 1950

Alan Turing, October 1950

«I propose to consider the question: 'Can machines think?」»

MIND

A QUARTERLY REVIEW

OF

PSYCHOLOGY AND PHILOSOPHY



I.—COMPUTING MACHINERY AND INTELLIGENCE

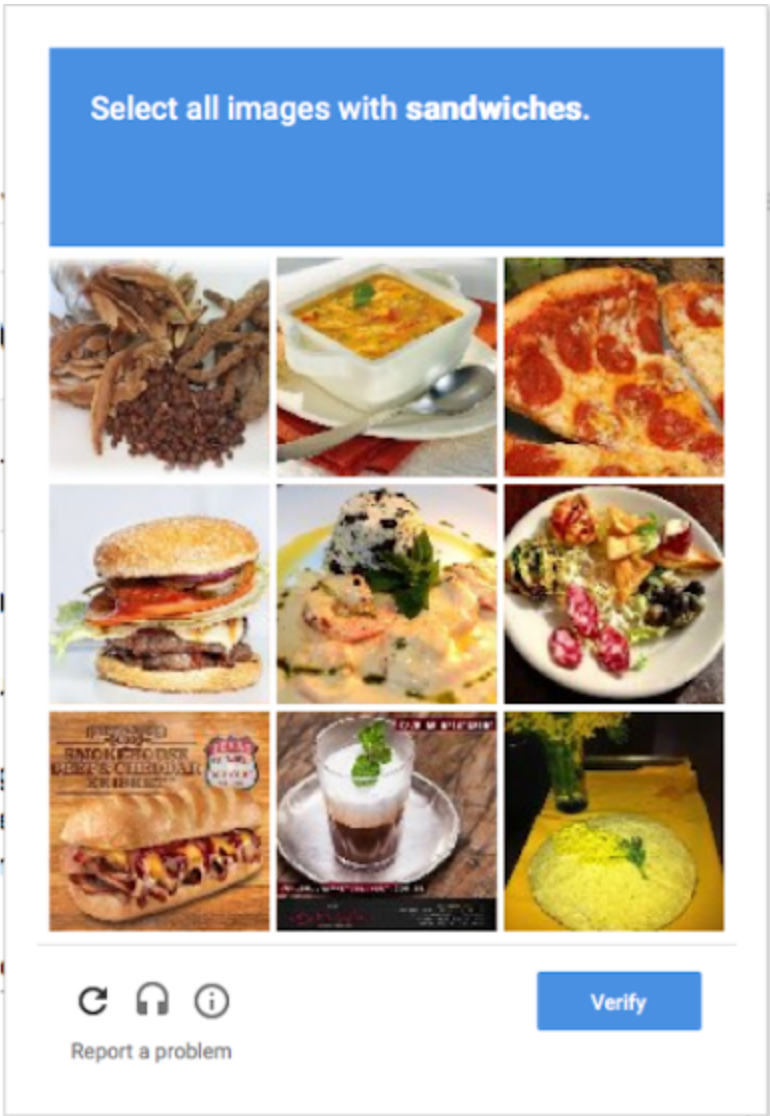
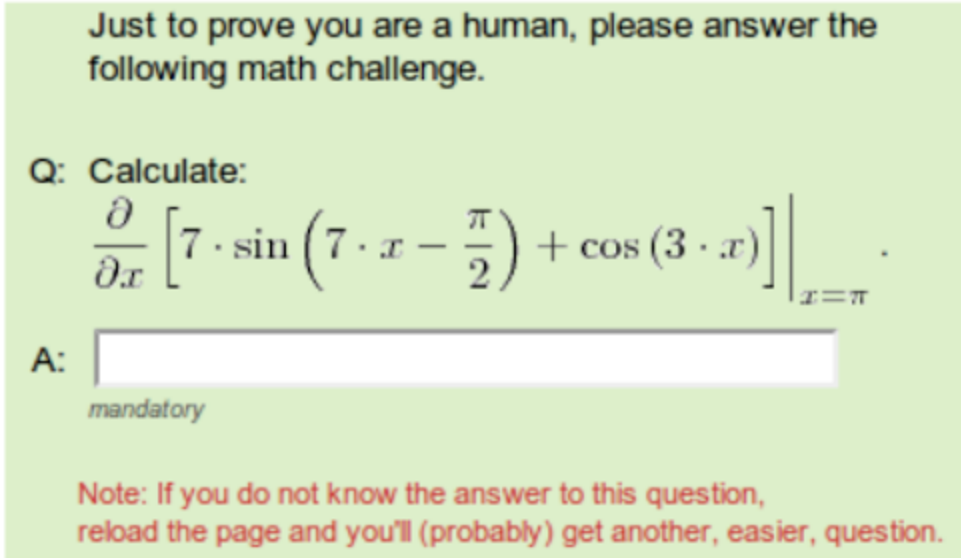
BY A. M. TURING

1. *The Imitation Game.*

I PROPOSE to consider the question, 'Can machines think?'

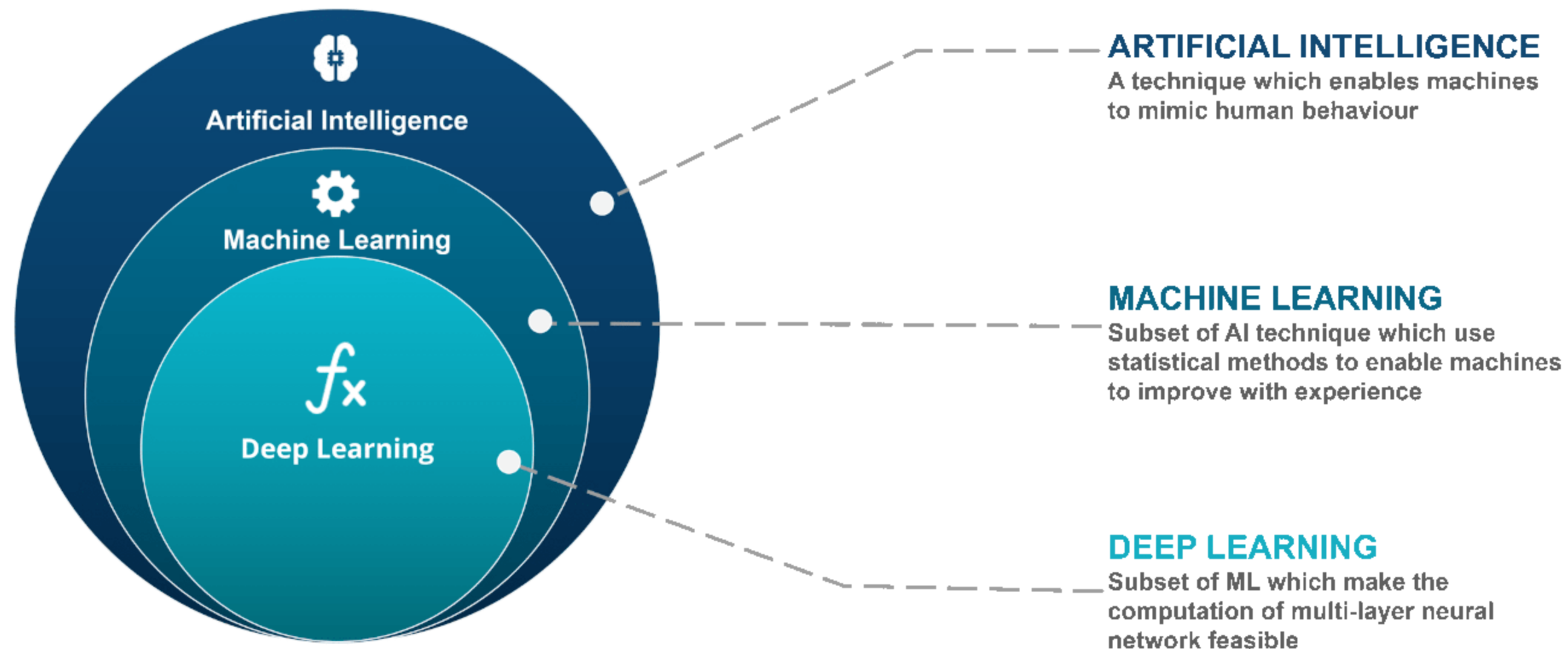
What is Machine Learning?

Turing Test



What is Machine Learning?

AI, Machine Learning, Deep Learning



What is Machine Learning?

What did people expect?

- «within ten years a digital computer will be the world's chess champion»,
«within ten years a digital computer will discover and prove an important new mathematical theorem.»
H. A. Simon and Allen Newell, 1958
- «machines will be capable, within twenty years, of doing any work a man can do.»
H. A. Simon, 1965
- «Within a generation ... the problem of creating 'artificial intelligence' will substantially be solved.»
Marvin Minsky, 1967
- «In from 3 to 8 years we will have a machine with the general intelligence of an average human being.»
Marvin Minsky (in Life Magazine), 1970

Applications

Applications

E-Commerce

- Recommender systems: «other people interested in this movie also watched...»
- Content personalization: show people content they probably like to see (can lead to filter bubbles)
- Chatbots: replace repetitive human work e.g. for simple customer support requests
- Dynamic pricing: how much is a customer willing to pay for the product (see airline tickets)
- Shoppable augmented reality—AR/VR: how would this table look in my living room

Applications

Healthcare

- Detection and diagnosis: e.g. via image or lab results analysis
- Treatment recommendation: prescript the correct action in the 80% of "easy" cases
- Discovery of new drugs: also supporting basic research (see e.g. AlphaFold)
- Epidemic prevention
- Streamlining of admin tasks: reduce costs and errors
- Patient risk identification

Applications

Finance

- Fraud detection: was this credit card really used by its rightful owner
- Credit scoring: will the lender be able to repay their credit?
- Insurance underwriting: sort for interesting opportunities to check
- Money laundering prevention: find connected/related accounts to track money transfers
- Portfolio management
- Algorithmic trading

Applications

Security

- Password protection and authentication: detect "stupid" passwords
- Cyber threat detection: automated observation and event analysis
- Phishing detection and prevention control: i.e. spam filter 2.0
- Vulnerability management: automated checks of source code for certain types

Applications

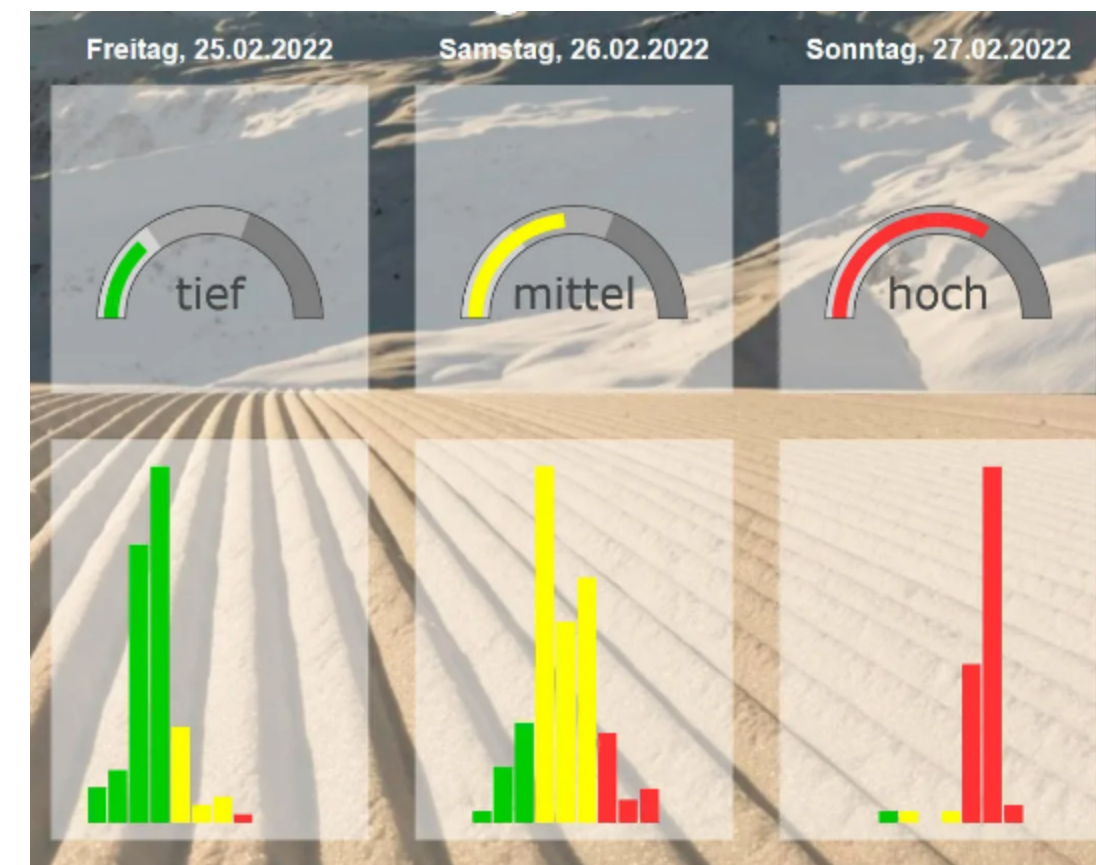
Real Estate

- Property valuation with hedonic models (linear regression)
- Rent and cost estimations
- Image Analysis (conditions, standards, view out the window etc.)

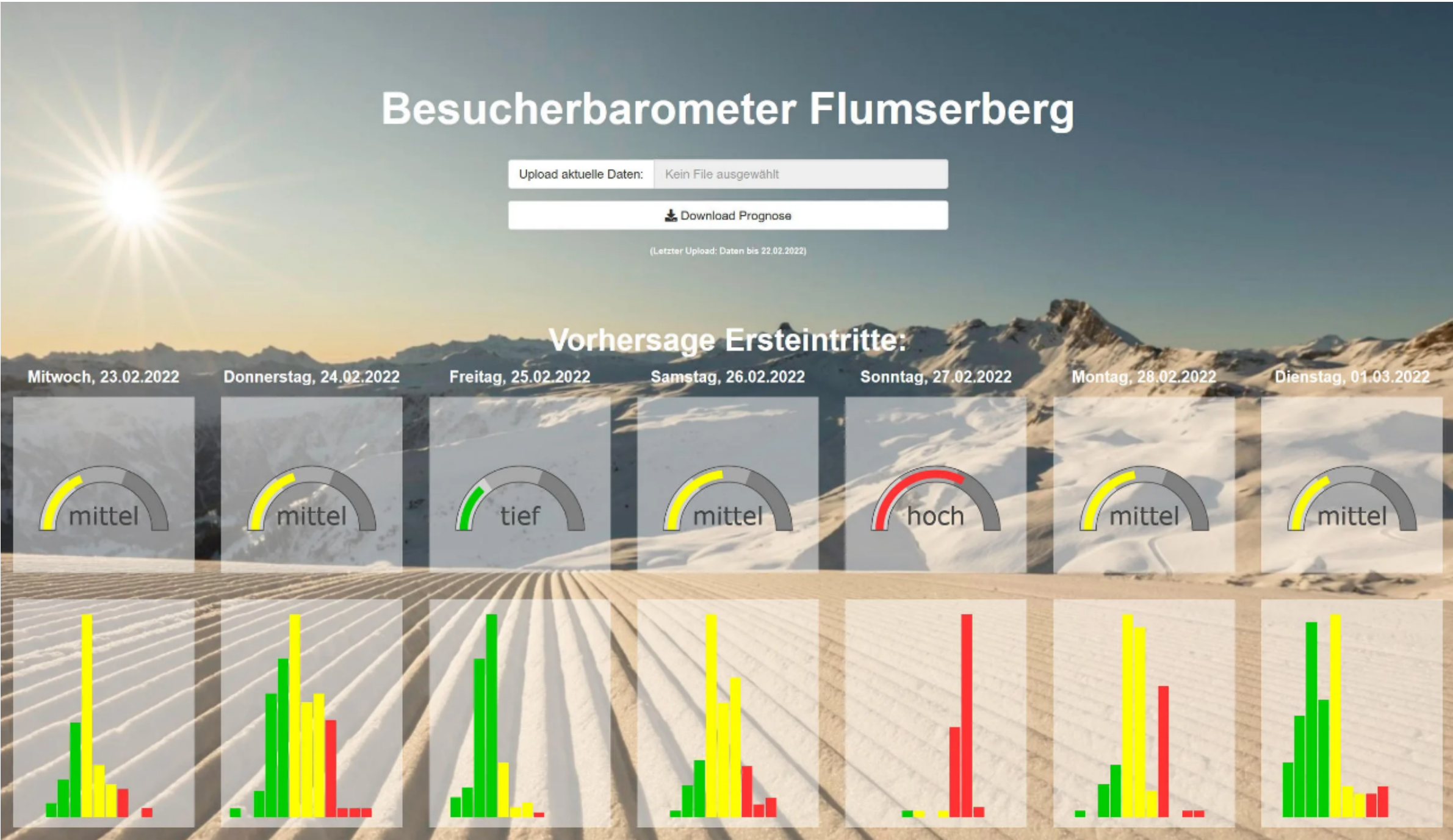
Applications

Visitor Prediction

- Goal: **Predict Number of Visitors**
- Modernes Vorhersage-Modell
- Used in Flumserberg since the 20/21 season
- How it is used:
 - Excel upload of current data
 - Prediction for the next days
 - Access via the web tool



Visitor Prediction



Applications

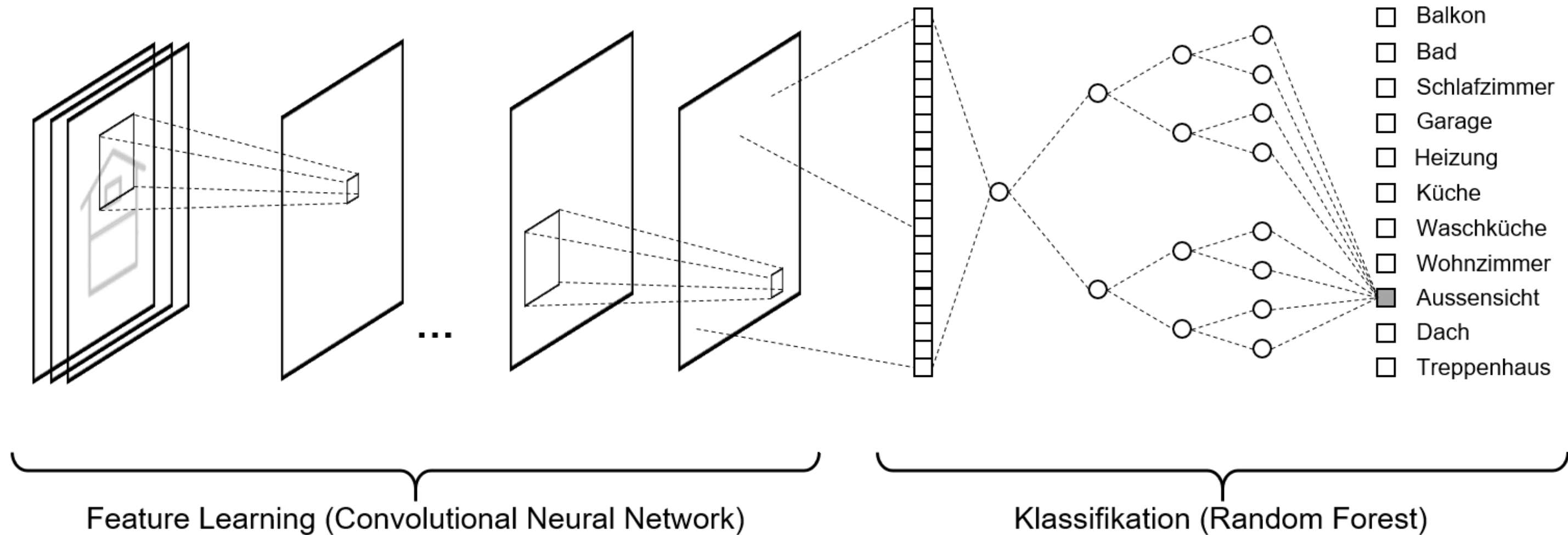
"ImmoEye": Project Overview

- Wüest-Innovation-Challenge
- Automatic **classification of content** in real estate images
- Additional **estimation of construction year** in kitchen, bathroom and outside-view images
- Frontend with image upload



Applications

"ImmoEye": Deep & Transfer Learning

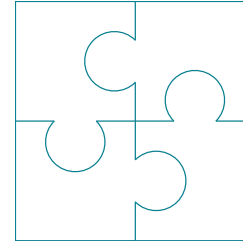


Applications

"ImmoEye": Used model

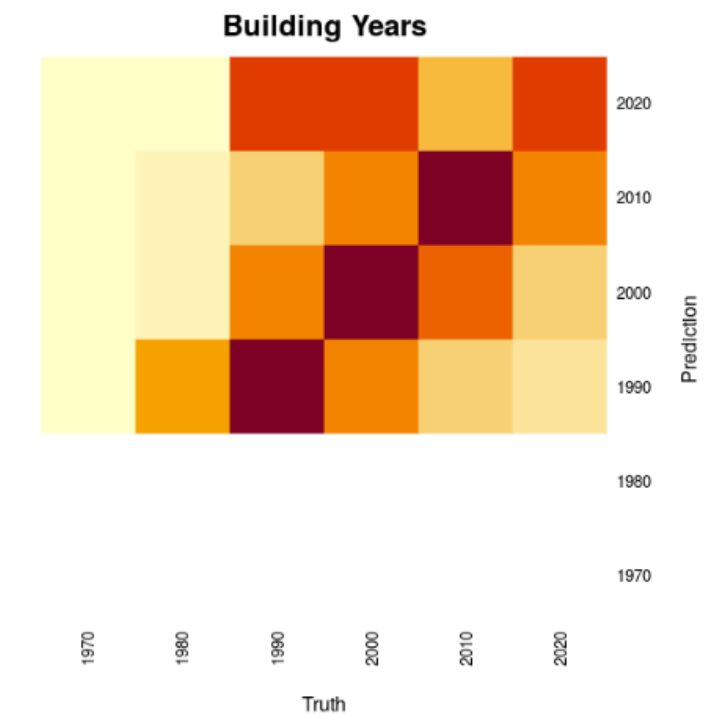
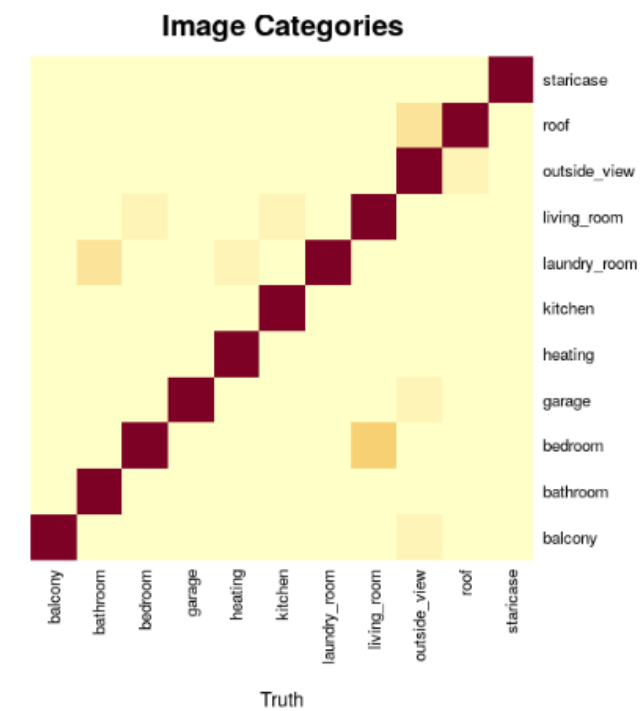
- Base model: "Mobilenet" architecture (Google)
- 1000 learned image categories (Imagenet)
- Transfer to our images via "Random Forest" classifier
- ~30'000 training images from Wüest Dimensions





"ImmoEye": Current performance

- Image classification:
 - 11 categories (roof, kitchen, etc.)
 - **>80% accuracy**
- Construction year kitchens:
 - 10-year categories
 - **~40% accuracy**



General Remarks

General Remarks

Model $\neq$ Reality

George E. P. Box:

«All models are wrong, but some are useful.»

So thinking a bit more about this this might be more useful to have a somewhat weaker model and understand its capabilities than having «the best» model without understanding its limitations.

General Remarks

Which model is best?

Unknown:

It depends!

Some applications (e.g. in the real estate industry) need external auditing. They will probably audit some coefficients for linear regression models but for sure the authorities do not (yet) know how to audit a "black box" neural network model.

General Remarks

More Complex $\neq$ Better

More complexity means more potential for mistakes

- Getting basic statistics wrong is easy enough
- Adding complex machine learning layers will not solve those mistakes but rather hide them better

Not all errors are equal

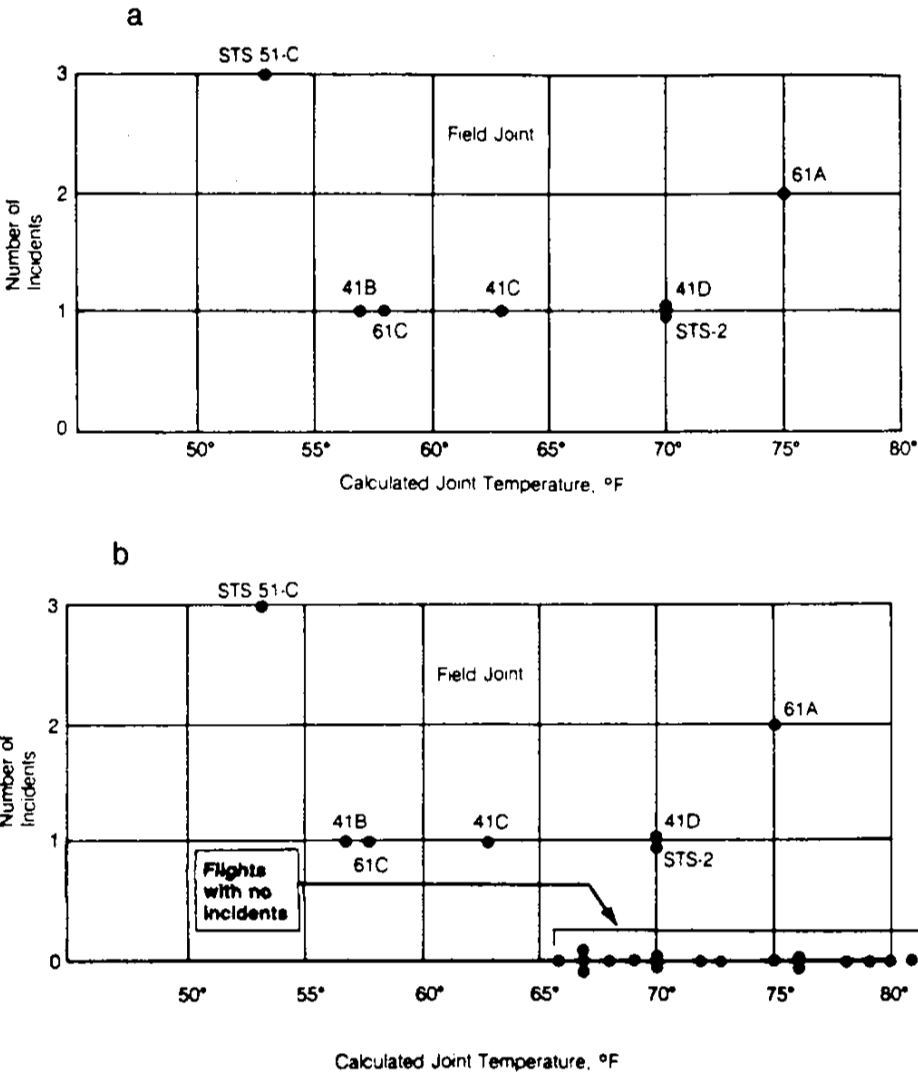
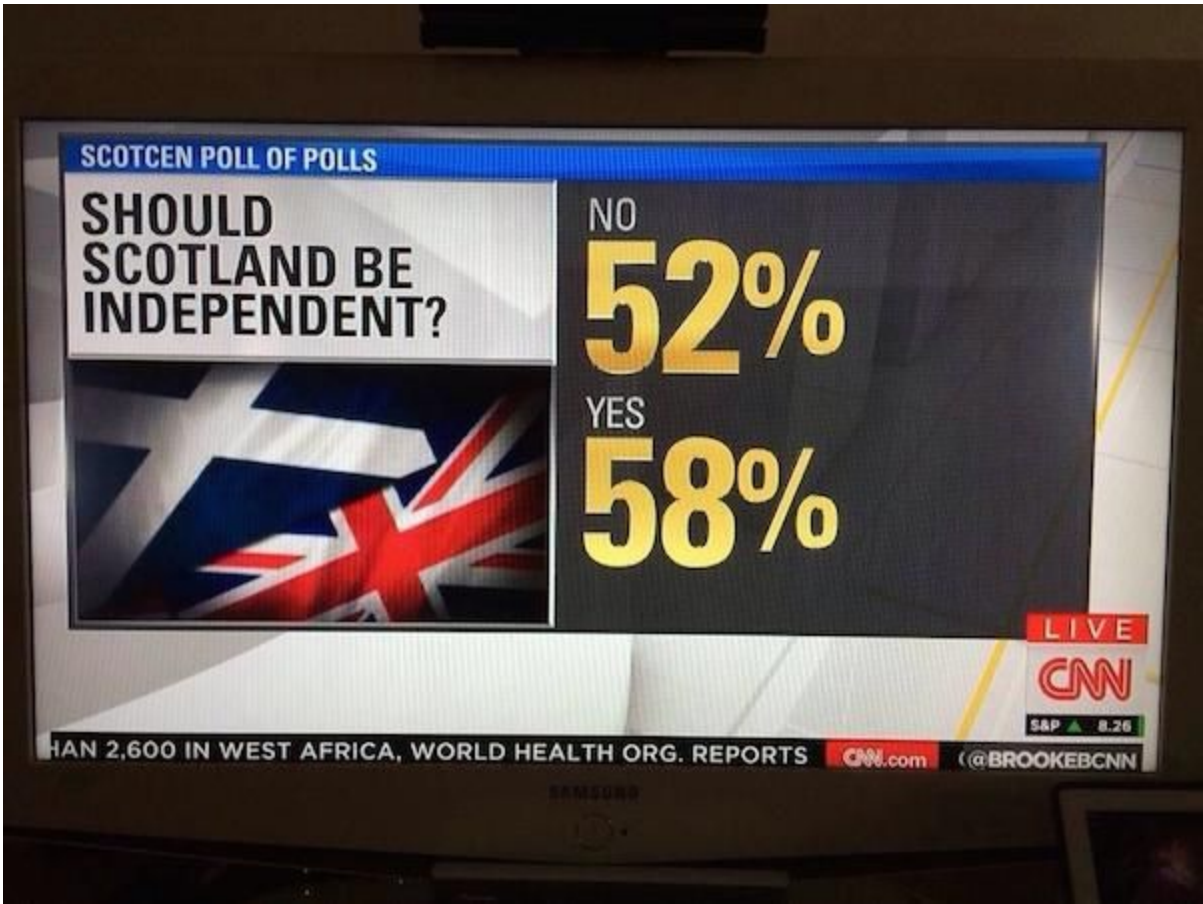
- Depending on the exact application the consequences of Type I and Type II errors may be vastly different

Variables/Records do have a real world Cost

- Non-parametric models will get better with more variables so it is very tempting just to pile up data
- Collecting and processing this data comes with real world costs
 - this is true both for more records as well as for more variables

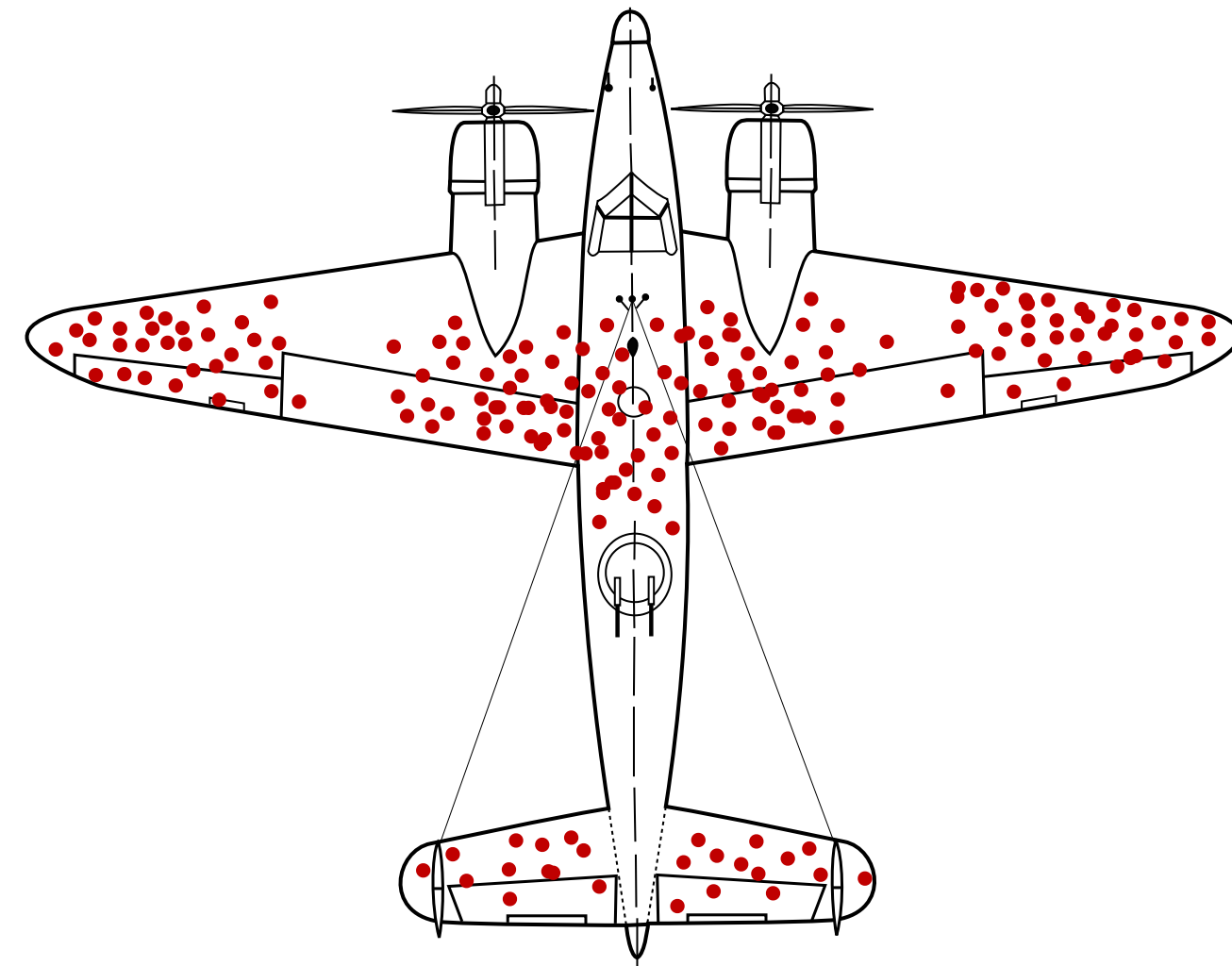
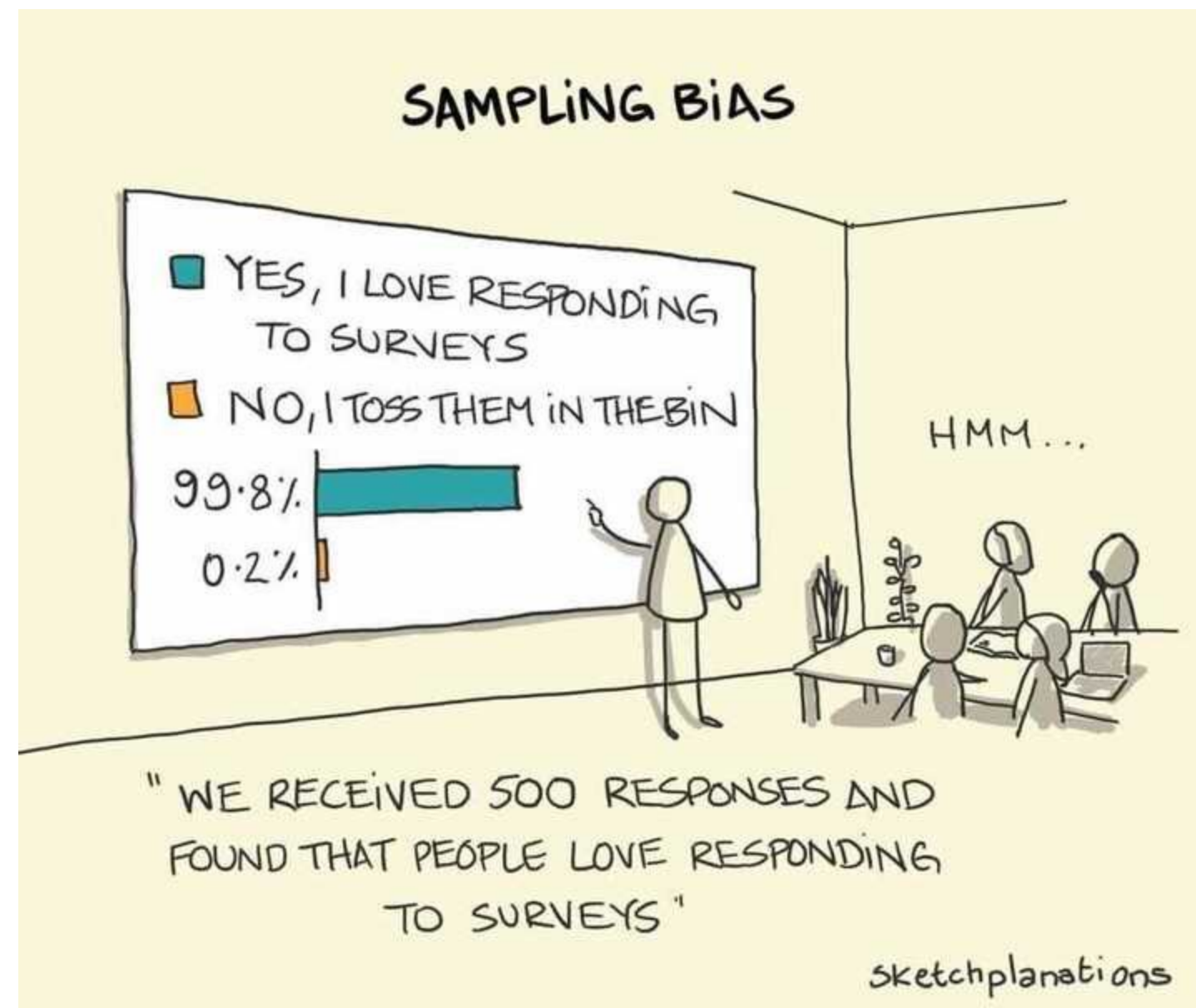
General Remarks

Complexity can hide "Trivial" Mistakes



General Remarks

Data is always Biased!





General Remarks

Context is Key

Sounds good to me.

Yes, I agree.

Sorry, I won't be there.

 Reply

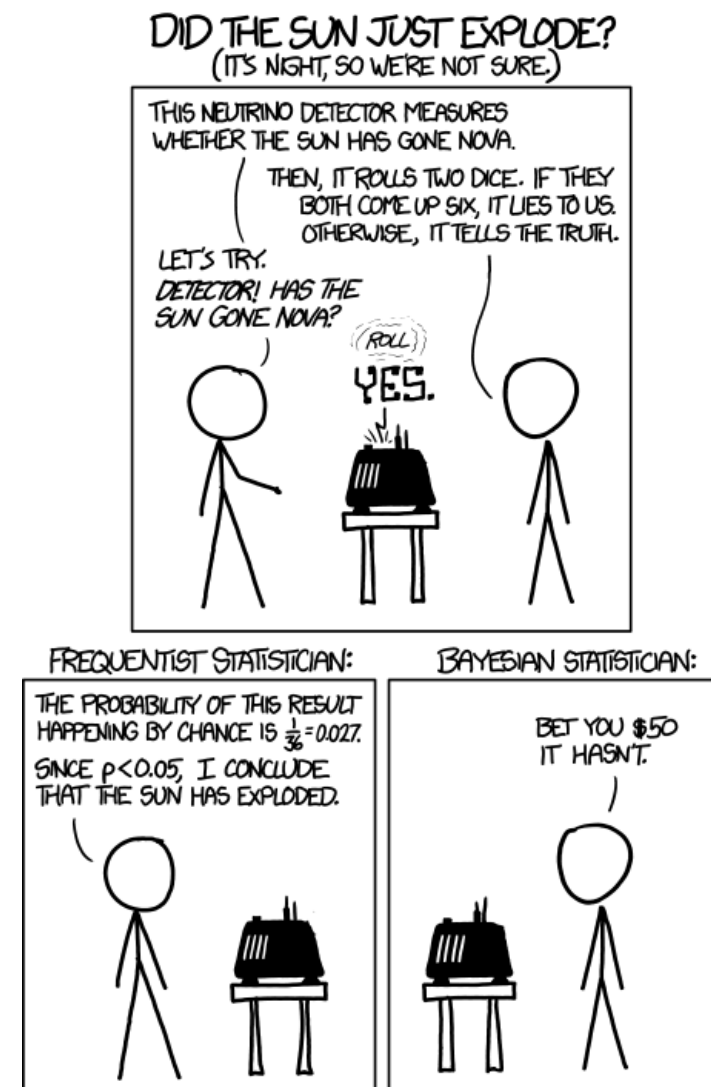
 Reply all

 Forward

General Remarks

Interpretation is Important

- No «stargazing»
- It's not only about p-values!



Ethical and Regulatory Considerations

Ethical and Regulatory Considerations

Privacy



Ethical and Regulatory Considerations

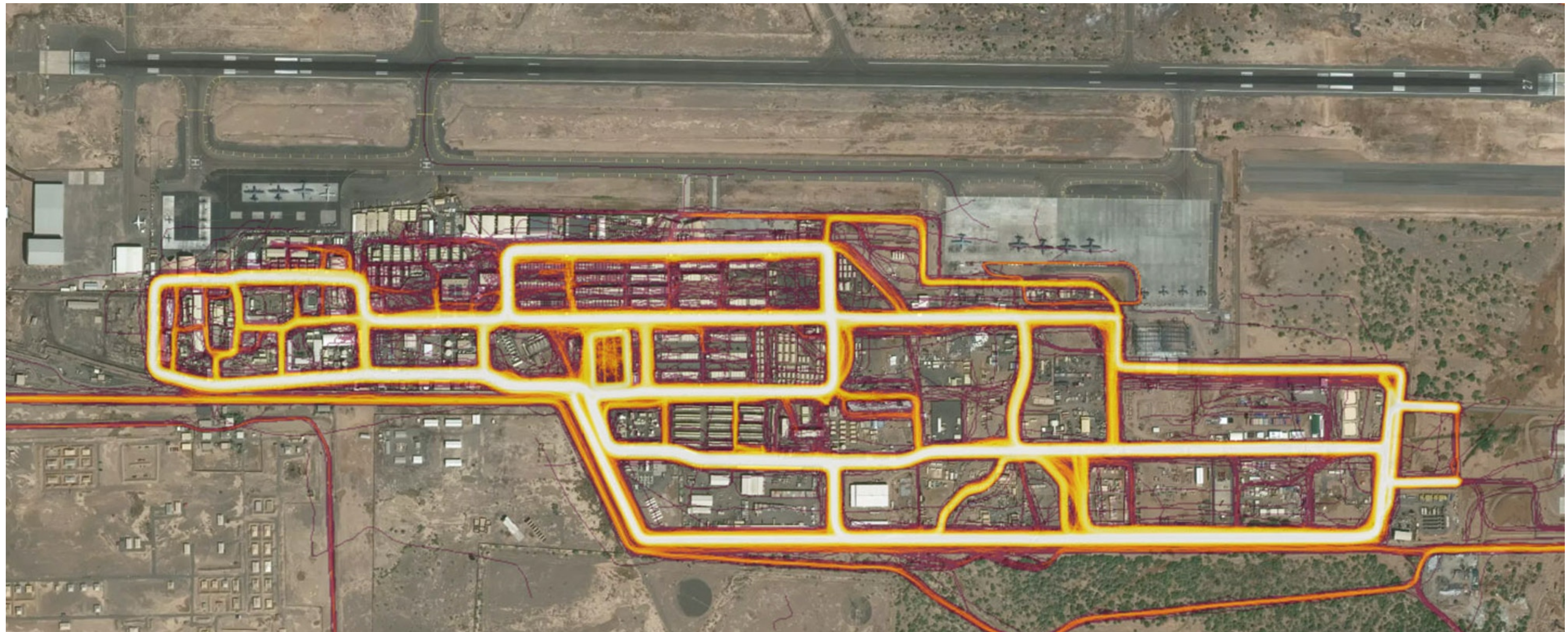
Data Privacy

- Large (top 10) US retailer
- Identified parents-to-be as good customers
- Built a model to predict pregnancy
- Sent vouchers for baby products
- Caused family problems and ensuing bad press
- Based on a [NYT article](#), details are controversial



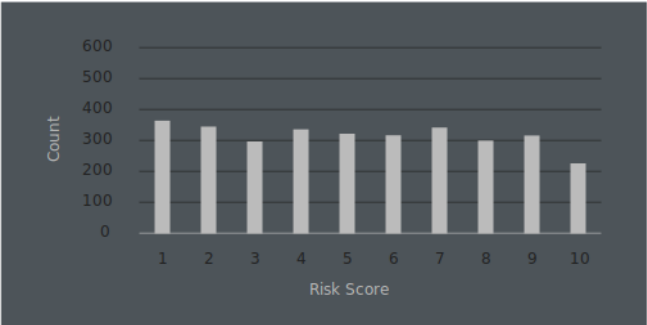
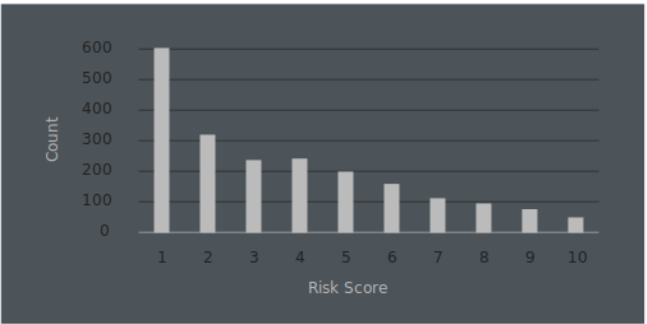
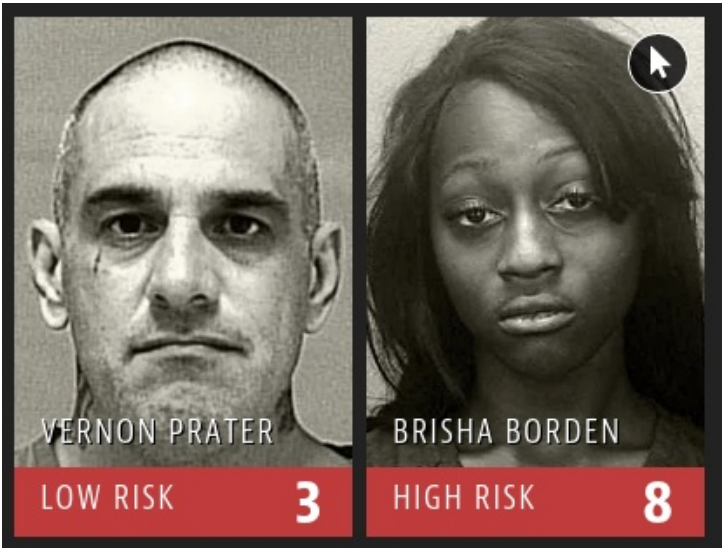
Ethical and Regulatory Considerations

Data Security



Ethical and Regulatory Considerations

Machine ≠ Unbiased



Source: <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

Daily Routine

Daily Routine

Machine Learning $\neq$ Model Tuning

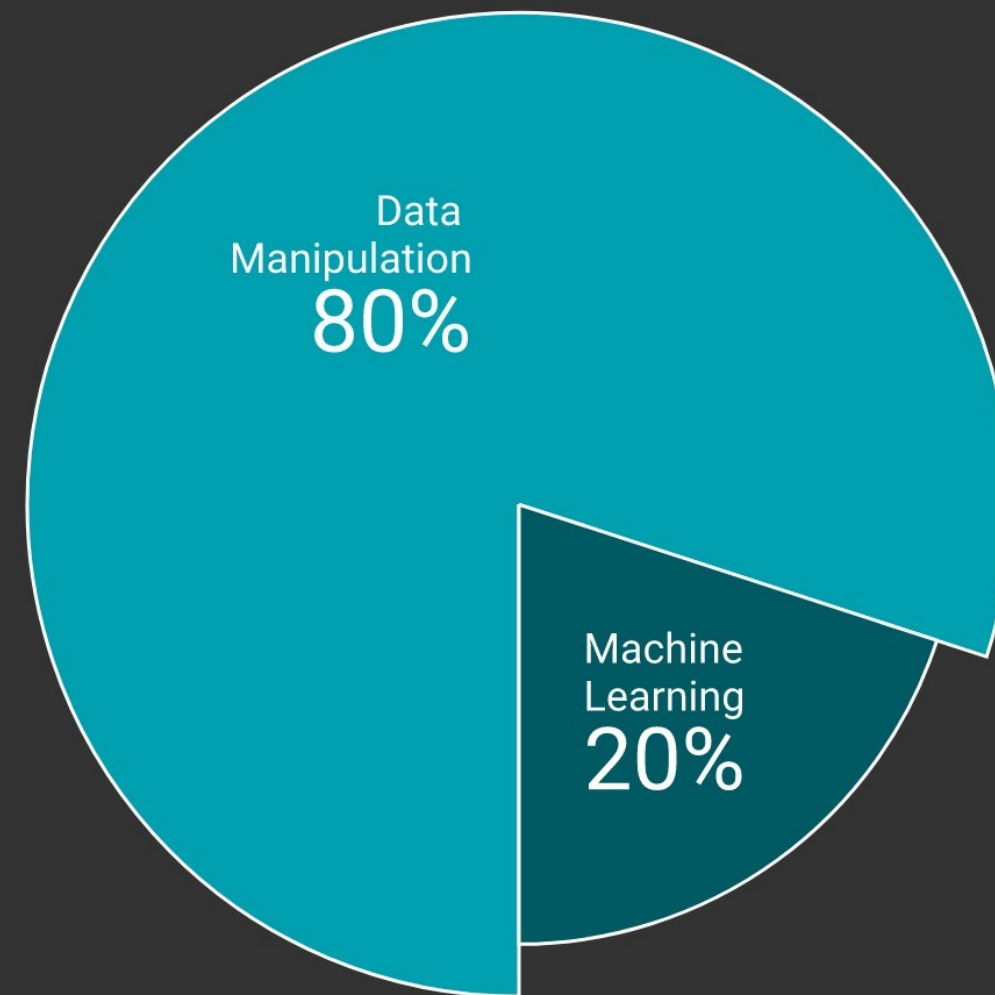
- Understanding business problems and business needs
- Find, extract, and prepare data
- Communications (business goals, data challenges, results etc.)
- Think about operationalisation early in the process
- Working in a team!



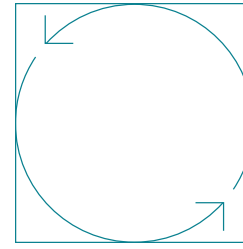
Machine Learning in Industry

Daily Routine

Time spent



Summary



Machine Learning in Industry

Summary

- Define the problem(s) and the requirements for the solution first
- Then choose the "best" model to solve this problem
 - the definition of "best" is very intricate and will depend on the business question
 - what is the task at hand, how expensive are additional data, different types of errors etc.
 - what model types is your team experienced in doing, what other resources are available
- **Never** do it the other way around («that is a very nice hammer, lets put in this screw with it»)
- Spend as much time on documentation and communication as you spend on parameter tuning!

October 30th, 2025

Thank you for your attention!

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